

Module -2

Step Response of First Order RL Circuits

Dr. Sathya P
Associate Professor
SENSE

Step response of an RL Circuit

$$i = i_t + i_{ss}$$

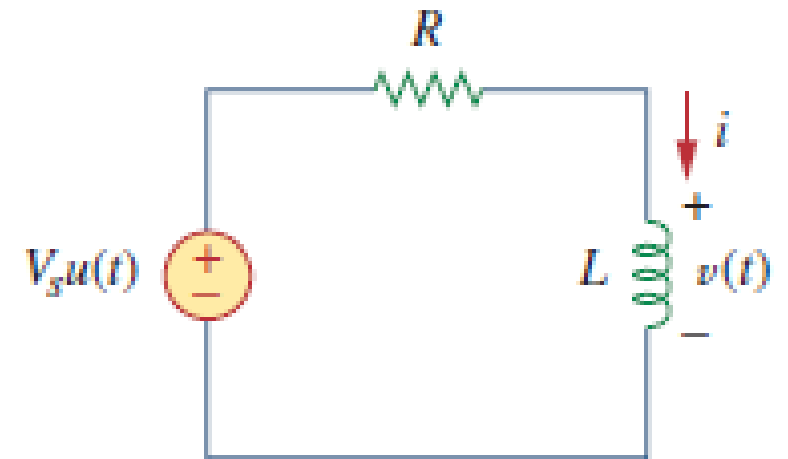
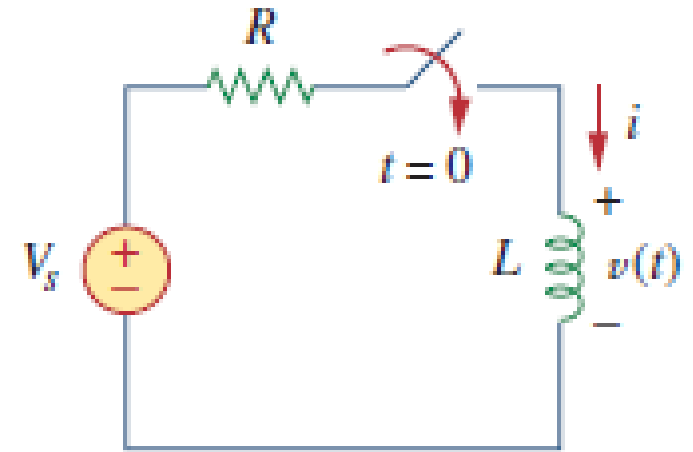
We know that the transient response is always a decaying exponential, that is,

$$i_t = Ae^{-t/\tau}, \quad \tau = \frac{L}{R}$$

where A is a constant to be determined.

$$i_{ss} = \frac{V_s}{R}$$

$$i = Ae^{-t/\tau} + \frac{V_s}{R}$$



$$i(0^+) = i(0^-) = I_0$$

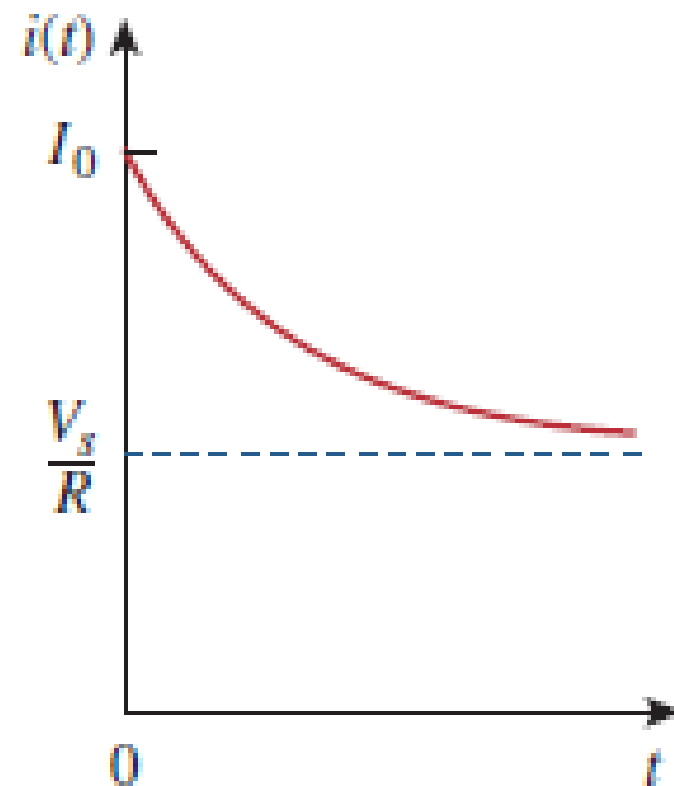
Thus, at $t = 0$,

$$I_0 = A + \frac{V_s}{R}$$

$$A = I_0 - \frac{V_s}{R}$$

$$i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R} \right) e^{-t/\tau}$$

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau}$$



Steps to solve:

1. Find the initial inductor current $i(0)$
2. Find the final inductor current $i(\infty)$
3. Find the time constant

$$i(t) = i(\infty) + [i(t_0) - i(\infty)]e^{-(t-t_0)/\tau}$$

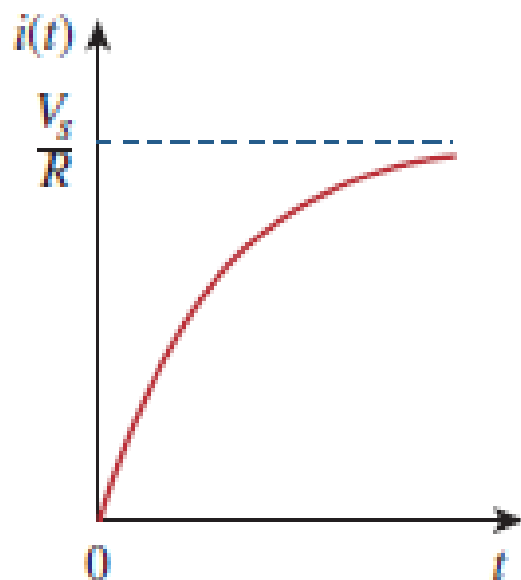
At $t = t_0$

$$i(t) = i(\infty) + [i(t_0) - i(\infty)]e^{-(t-t_0)/\tau}$$

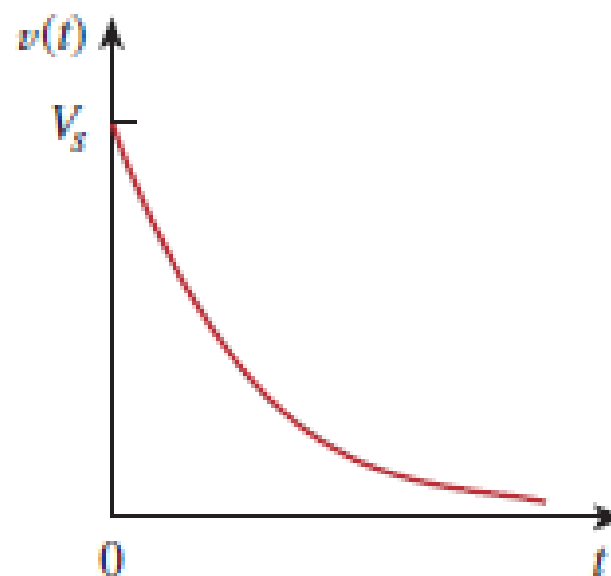
If $I_0 = 0$, then

$$i(t) = \begin{cases} 0, & t < 0 \\ \frac{V_s}{R}(1 - e^{-t/\tau}), & t > 0 \end{cases} \quad i(t) = \frac{V_s}{R}(1 - e^{-t/\tau})u(t)$$

$$v(t) = L \frac{di}{dt} = V_s \frac{L}{\tau R} e^{-t/\tau}, \quad \tau = \frac{L}{R}, \quad t > 0$$



(a)

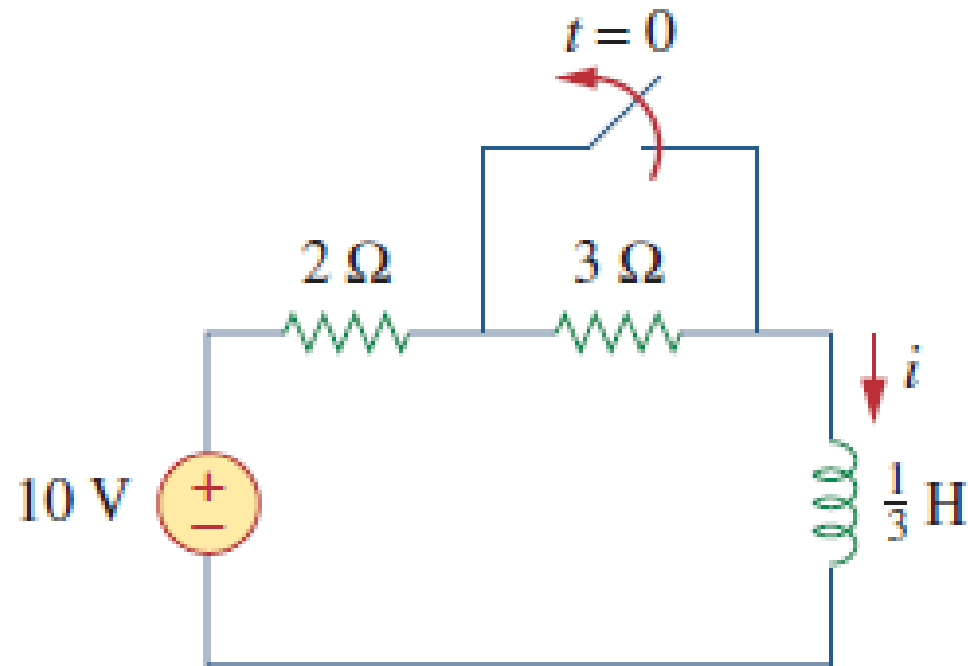


(b)

Step responses of an RL circuit with no initial inductor current: (a) current response, (b) voltage response.

Numerical - 1

Determine $i(t)$ for $t > 0$. Assume that the switch has been closed for a long time.



Solution:

When $t < 0$, the $3\text{-}\Omega$ resistor is short-circuited, and the inductor acts like a short circuit. The current through the inductor at $t = 0^-$ (i.e., just before $t = 0$) is

$$i(0^-) = \frac{10}{2} = 5 \text{ A}$$

Since the inductor current cannot change instantaneously,

$$i(0) = i(0^+) = i(0^-) = 5 \text{ A}$$

When $t > 0$, the switch is open. The $2\text{-}\Omega$ and $3\text{-}\Omega$ resistors are in series, so that

$$i(\infty) = \frac{10}{2 + 3} = 2 \text{ A}$$

The Thevenin resistance across the inductor terminals is

$$R_{\text{Th}} = 2 + 3 = 5 \, \Omega$$

For the time constant,

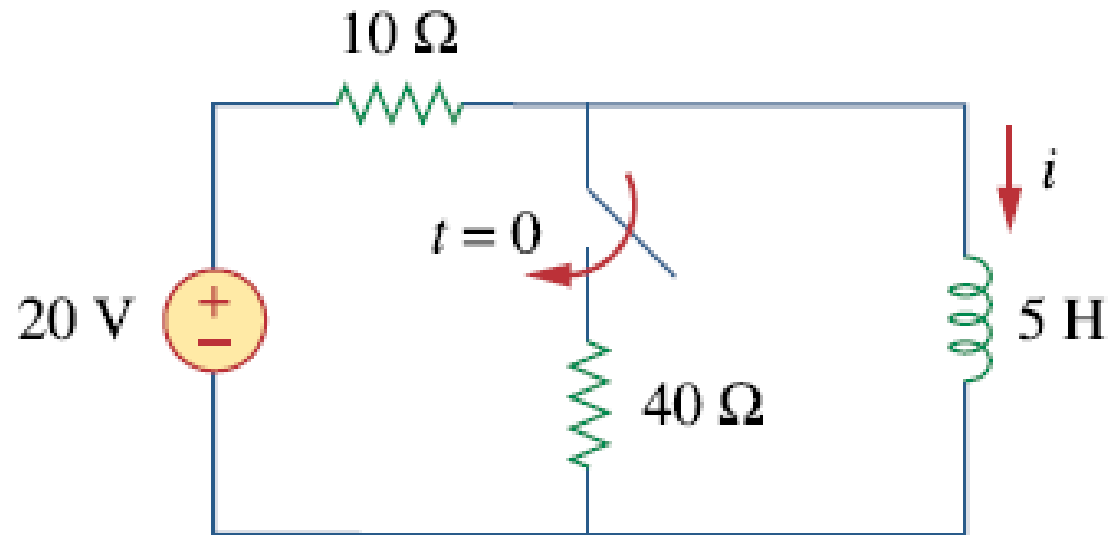
$$\tau = \frac{L}{R_{\text{Th}}} = \frac{\frac{1}{3}}{5} = \frac{1}{15} \, \text{s}$$

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau}$$

$$= 2 + (5 - 2)e^{-15t} = 2 + 3e^{-15t} \, \text{A}, \quad t > 0$$

Numerical - 2

Determine $i(t)$ for $t > 0$.



Solution:

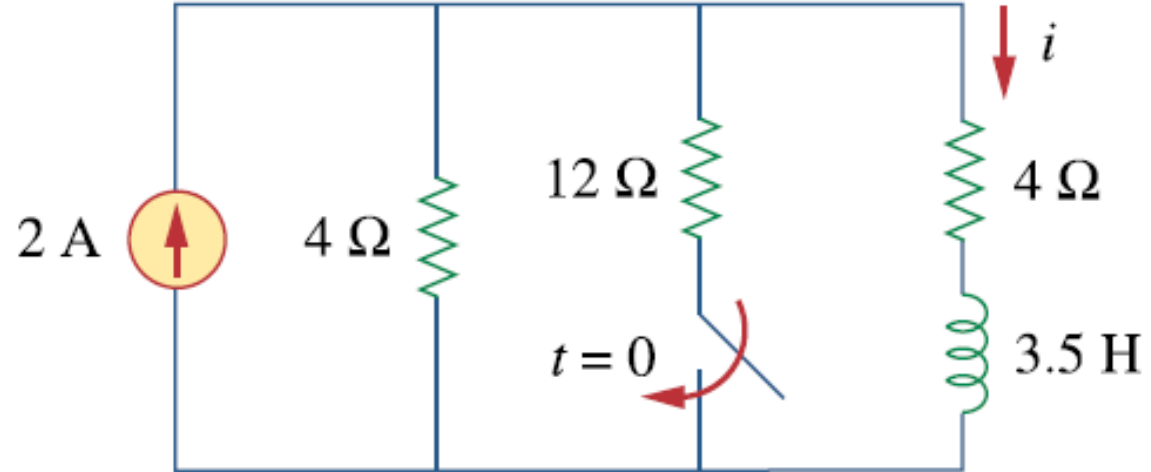
$$i(0) = \frac{20}{10} = 2 \text{ A}, \quad i(\infty) = 2 \text{ A}$$

$$i(t) = i(\infty) + [i(0) - i(\infty)] e^{-t/\tau}$$

$$i(t) = \underline{\underline{2 \text{ A}}}$$

Numerical - 3

Determine $i(t)$ for both $t > 0$ and $t < 0$.



- Solution:

Before $t = 0$, i is obtained by current division or

$$i(t) = \frac{4}{4+4} (2) = \underline{1 \text{ A}}$$

After $t = 0$,

$$i(t) = i(\infty) + [i(0) - i(\infty)] e^{-t/\tau}$$

$$\tau = \frac{L}{R_{eq}}, \quad R_{eq} = 4 + 4 \parallel 12 = 7 \Omega$$

$$\tau = \frac{3.5}{7} = \frac{1}{2}$$

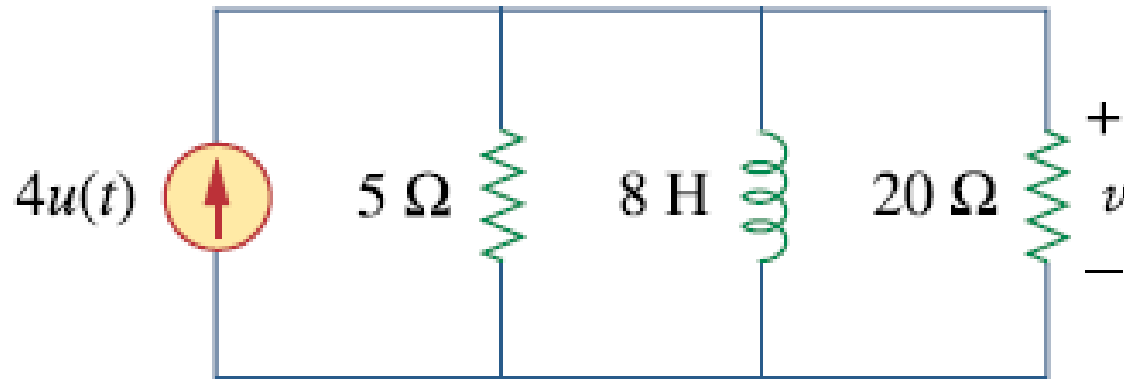
$$i(0) = 1, \quad i(\infty) = \frac{4 \parallel 12}{4 + 4 \parallel 12} (2) = \frac{3}{4+3} (2) = \frac{6}{7}$$

$$i(t) = \frac{6}{7} + \left(1 - \frac{6}{7}\right) e^{-2t}$$

$$i(t) = \underline{\frac{1}{7} (6 + e^{-2t}) \text{ A}}$$

Numerical - 4

Find $v(t)$ for $t > 0$ if the initial inductor current is zero.



- Solution:

Let i be the inductor current.

$$\text{For } t < 0, \quad u(t) = 0 \quad \longrightarrow \quad i(0) = 0$$

$$\text{For } t > 0, \quad R_{\text{eq}} = 5 \parallel 20 = 4 \, \Omega, \quad \tau = \frac{L}{R_{\text{eq}}} = \frac{8}{4} = 2$$

$$i(\infty) = 4$$

$$i(t) = i(\infty) + [i(0) - i(\infty)] e^{-t/\tau}$$

$$i(t) = 4(1 - e^{-t/2})$$

$$v(t) = L \frac{di}{dt} = (8)(-4) \left(\frac{-1}{2} \right) e^{-t/2}$$

$$v(t) = \underline{16 e^{-0.5t} \text{ V}}$$